2.15 Hyperbolic Functions: Domain and Range

Worked Solutions

1 a)

$y = \sinh x$ is one-to-one

1 b)

$y = \cosh x$ is many-to-one

1 c)

$y = \tanh x$ is one-to-one

2 a)

$$y = \sinh x$$

Domain: $\{x : x \in \mathbb{R}\}$

Range: $\{y : y \in \mathbb{R}\}$

2 b)

$$y = \cosh x$$

Domain: $\{x : x \in \mathbb{R}\}$

Range: $\{y : y \in \mathbb{R},\ y \geq 1\}$

2 c)

$$y = \tanh x$$

Domain: $\{x : x \in \mathbb{R}\}$

Range: $\{y : y \in \mathbb{R},\ -1 < y < 1\}$

3 a)

$$y = \sinh x$$

Range: $\{y : y \in \mathbb{R},\ y < 0\}$

3 b)

$$y = \cosh x$$

Range: $\{y : y \in \mathbb{R},\ y \geq 1\}$

3 c)

$$y = \tanh x$$

$$\text{Range: } \{y : y \in \mathbb{R},\ -1 < y < 0\}$$

4 a)

$$\text{Range: } \{f(x) : f(x) \in \mathbb{R},\ f(x) \geq 0\}$$

4 b)

$$\text{Range: } \{y : y \in \mathbb{R},\ y \geq 4\}$$

4 c)

$$\text{Range: } \{g(x) : g(x) \in \mathbb{R},\ -1 < g(x) < 1\}$$

4 d)

$$\text{Range: } \{y : y \in \mathbb{R}\}$$

4 e)

$$\text{Range: } \{h(x) : h(x) \in \mathbb{R},\ h(x) \leq -1\}$$

4 f)

$$\text{Range: } \{y : y \in \mathbb{R},\ 5 \leq y < 6\}$$

5 a)

$$\begin{aligned} &2\sinh x + 4\cosh x \\ &\equiv 2 \times \frac{e^x - e^{-x}}{2} + 4 \times \frac{e^x + e^{-x}}{2} \\ &\equiv e^x - e^{-x} + 2e^x + 2e^{-x} \\ &\equiv 3e^x + e^{-x} \end{aligned}$$

5 b)

$3e^x > 0$ for all $x \in \mathbb{R}$

So $3e^x + e^{-x} > e^{-x}$

$\therefore 2\sinh x + 4\cosh x > e^{-x}$

6)

$$
\begin{aligned}
&4\sinh^2 x - e^{-2x} \\
&\equiv 4\left(\frac{e^x - e^{-x}}{2}\right)^2 - e^{-2x} \\
&\equiv (e^x - e^{-x})^2 - e^{-2x} \\
&\equiv e^{2x} - 2 + e^{-2x} - e^{-2x} \\
&\equiv e^{2x} - 2
\end{aligned}
$$

$e^{2x} > 0$ for all $x \in \mathbb{R}$

So $e^{2x} - 2 > -2$

$\therefore 4\sinh^2 x - e^{-2x} > -2$

7 a)

$$\sinh 2x \equiv 2\sinh x \cosh x$$

7 b i)

$$\{y : y \in \mathbb{R},\ y > 0\}$$

7 b ii)

$$\{y : y \in \mathbb{R},\ y > 1\}$$

7 c)

$$\sinh 2x = 2\sinh x \cosh x$$

From b) i) we know that $\sinh x > 0$ and from b) ii) we know that $\cosh x > 1$

So $2\cosh x > 2$

Multiply both sides by $\sinh x$ (which maintains the inequality as $\sinh x > 0$):

$$2\sinh x \cosh x > 2\sinh x$$

Also we know that $2\sinh x > \sinh x$ as $\sinh x > 0$

$$\therefore \sinh 2x = 2\sinh x \cosh x > 2\sinh x > \sinh x$$

$$\Rightarrow \sinh 2x > \sinh x \text{ for } x > 0$$

8)

$$f(x) = \sinh x + 2\cosh x$$
$$f'(x) = \cosh x + 2\sinh x$$

Stationary points exist when $f'(x) = 0$

$$\Rightarrow \cosh x + 2\sinh x = 0 \Rightarrow 1 + 2\tanh x = 0 \Rightarrow \tanh x = -\frac{1}{2}$$

$$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+(-\frac{1}{2})}{1-(-\frac{1}{2})}\right) = \frac{1}{2}\ln\left(\frac{\frac{1}{2}}{\frac{3}{2}}\right) = \frac{1}{2}\ln\frac{1}{3}$$

$$f\left(\frac{1}{2}\ln\frac{1}{3}\right) = \sinh\left(\frac{1}{2}\ln\frac{1}{3}\right) + 2\cosh\left(\frac{1}{2}\ln\frac{1}{3}\right)$$
$$= \frac{e^{\frac{1}{2}\ln\frac{1}{3}} - e^{-\frac{1}{2}\ln\frac{1}{3}}}{2} + 2\times\frac{e^{\frac{1}{2}\ln\frac{1}{3}} + e^{-\frac{1}{2}\ln\frac{1}{3}}}{2}$$
$$= \frac{e^{\ln(\frac{1}{\sqrt{3}})} - e^{\ln(\sqrt{3})}}{2} + e^{\ln(\frac{1}{\sqrt{3}})} + e^{\ln(\sqrt{3})}$$
$$= \frac{3}{2}e^{\ln(\frac{1}{\sqrt{3}})} + \frac{1}{2}e^{\ln(\sqrt{3})}$$
$$= \frac{3}{2\sqrt{3}} + \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \sqrt{3}$$

Range is $\{f(x) : f(x) \in \mathbb{R},\ f(x) \geq \sqrt{3}\}$

9)

$$f(x) = 3\cosh x - \sinh x\cosh x$$
$$f'(x) = 3\sinh x - \sinh^2 x - \cosh^2 x$$

Stationary points exist when $f'(x) = 0$

$$\Rightarrow 3\sinh x - \sinh^2 x - \cosh^2 x = 0$$

$$\Rightarrow 3\sinh x - \sinh^2 x - (\sinh^2 x + 1) = 0$$
$$\Rightarrow 0 = 2\sinh^2 x - 3\sinh x + 1$$
$$\Rightarrow 0 = (2\sinh x - 1)(\sinh x - 1)$$
$$\sinh x = \frac{1}{2} \text{ or } \sinh x = 1$$

$$\sinh x = \frac{1}{2} \Rightarrow x = \ln\left(\frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 + 1}\right) = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$$

$$\sinh x = 1 \Rightarrow x = \ln\left(1 + \sqrt{1^2 + 1}\right) = \ln(1+\sqrt{2})$$

When $x = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$:

$$f\left(\ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)\right) = 3\cosh\left(\ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)\right) - \sinh\left(\ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)\right)\cosh\left(\ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)\right)$$

$$= 3 \times \frac{e^{\ln(\frac{1}{2}+\frac{\sqrt{5}}{2})} + e^{-\ln(\frac{1}{2}+\frac{\sqrt{5}}{2})}}{2} - \frac{e^{\ln(\frac{1}{2}+\frac{\sqrt{5}}{2})} - e^{-\ln(\frac{1}{2}+\frac{\sqrt{5}}{2})}}{2} \times \frac{e^{\ln(\frac{1}{2}+\frac{\sqrt{5}}{2})} + e^{-\ln(\frac{1}{2}+\frac{\sqrt{5}}{2})}}{2}$$

$$= 3 \times \frac{\frac{1}{2}+\frac{\sqrt{5}}{2}+\frac{\sqrt{5}}{2}-\frac{1}{2}}{2} - \frac{\frac{1}{2}+\frac{\sqrt{5}}{2}-(\frac{\sqrt{5}}{2}-\frac{1}{2})}{2} \times \frac{\frac{1}{2}+\frac{\sqrt{5}}{2}+\frac{\sqrt{5}}{2}-\frac{1}{2}}{2}$$

$$= \frac{3\sqrt{5}}{2} - \frac{1}{2} \times \frac{\sqrt{5}}{2} = \frac{3\sqrt{5}}{2} - \frac{\sqrt{5}}{4} = \frac{5\sqrt{5}}{4} \approx 2.795$$

When $x = \ln(1+\sqrt{2})$:

$$f(\ln(1+\sqrt{2})) = 3\cosh(\ln(1+\sqrt{2})) - \sinh(\ln(1+\sqrt{2}))\cosh(\ln(1+\sqrt{2}))$$

$$= 3 \times \frac{e^{\ln(1+\sqrt{2})} + e^{-\ln(1+\sqrt{2})}}{2} - \frac{e^{\ln(1+\sqrt{2})} - e^{-\ln(1+\sqrt{2})}}{2} \times \frac{e^{\ln(1+\sqrt{2})} + e^{-\ln(1+\sqrt{2})}}{2}$$

$$= 3 \times \frac{1+\sqrt{2}+\sqrt{2}-1}{2} - \frac{1+\sqrt{2}-(\sqrt{2}-1)}{2} \times \frac{1+\sqrt{2}+\sqrt{2}-1}{2}$$

$$= 3\sqrt{2} - \sqrt{2} = 2\sqrt{2} \approx 2.828$$

Range is $f(x) \in \left(\frac{5\sqrt{5}}{4}, 2\sqrt{2}\right]$

10 a)

$$f(x) = \frac{e^{2x}+e^{-2x}}{(e^x+e^{-x})^2} - \tanh^2 x$$

$$\equiv \frac{e^{2x}+e^{-2x}}{(e^x+e^{-x})^2} - \frac{\sinh^2 x}{\cosh^2 x}$$

$$\equiv \frac{e^{2x}+e^{-2x}}{(e^x+e^{-x})^2} - \frac{(\frac{e^x-e^{-x}}{2})^2}{(\frac{e^x+e^{-x}}{2})^2}$$

$$\equiv \frac{e^{2x}+e^{-2x}}{(e^x+e^{-x})^2} - \frac{(e^x-e^{-x})^2}{(e^x+e^{-x})^2}$$

$$\equiv \frac{e^{2x}+e^{-2x}}{(e^x+e^{-x})^2} - \frac{e^{2x}-2+e^{-2x}}{(e^x+e^{-x})^2}$$

$$\equiv \frac{e^{2x}+e^{-2x}-e^{2x}+2-e^{-2x}}{(e^x+e^{-x})^2}$$

$$\equiv \frac{2}{(e^x+e^{-x})^2}$$

$$\equiv 2(e^x+e^{-x})^{-2}$$

10 b)

Stationary points exist when $f'(x) = 0$

$$f'(x) = -4(e^x-e^{-x})(e^x+e^{-x})^{-3}$$

$$\equiv -\frac{4(e^x-e^{-x})}{(e^x+e^{-x})^3}$$

$$f'(x) = 0 \Rightarrow -\frac{4(e^x-e^{-x})}{(e^x+e^{-x})^3} = 0$$

$$\Rightarrow e^x - e^{-x} = 0$$

$$\Rightarrow e^{2x} - 1 = 0$$
$$\Rightarrow e^{2x} = 1$$
$$\Rightarrow 2x = 0$$
$$\Rightarrow x = 0$$
$$f(0) = 2(e^0 + e^0)^{-2} = \frac{1}{2}$$

So the stationary point has coordinates $\left(0, \frac{1}{2}\right)$

10 c)

As $x \to +\infty$,

$$\frac{2}{(e^x + e^{-x})^2} \to 0$$

As $x \to -\infty$,

$$\frac{2}{(e^x + e^{-x})^2} \to 0$$

So the range of f is $\{f(x) : f(x) \in \mathbb{R},\ 0 < f(x) \leq \frac{1}{2}\}$